AI Powered Disability Discrimination: How Do You Lip Read a Robot Recruiter?

By Susan Scott-Parker

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00:00
Hello, everyone. I'm Susan Scott Parker from Business Disability International. And before I start, I'd just like to share a little film that we made, animation. Just to give you some basic insights into the conversation we're having this morning. Thank you.

00:26
Imagine if everyone in your life who has a disability suddenly just disappeared. Oops there go the 10% dyslexic, the 13% hearing impaired, 10% with sight loss, diabetes, arthritis, learning disabilities, depression, facial disfigurement, dementia and stroke, asthma, speech impairment, wheelchair users, migraines, anxiety, mutations and brain damage, mucular dystrophy. Before we know it, there'd be no one left. There are more than 1 billion people with disabilities worldwide, and the number is growing. Yet how many of us when we hear this word disability, automatically click to the old assumptions. Nothing to do with me. It's just a few wheelchair users, blind people - dogs, doctors and charities take care of them. When actually the experience of disability, the unfairness and disadvantage that it brings is very much about me, in fact, about all of us, about our experiences we go through life, and about getting older, which most of us would like to do. Actually, we all know human perfection is rare. Why bother with the Olympics if everyone can do a three minute mile? The real problem is our sad tendency to make assumptions about each other on the basis of labels. How we ourselves expect to be treated is changing. As we recognize that we all have a basic right to respect, dignity and choice. We need to replace the unacceptable old focus on medical labels with the new focus on individual potential, dignity, and human rights. The old brain makes assumptions. Everyone just knows that blind people can't use the internet. The new brain asks, how could this talented blind researcher do this job if we were clever? Simply put, impairment is what happens to you, disability is what we do to you. Refuse the ramp. Refuse flexitime. Insist you apply online even though the technology won't let you. Refuse to let you use a different keyboard. Assume you're just not worth the bother. That old click comes through especially loud and clear when you hear the traditional challenge. So what is the business case for hiring disabled people? But just think, nobody asks, what is the business case for hiring, say, Canadians? Everyone knows some can do the job, some can't. Some have the talent, some don't. Some need adjustments, some do not. We don't generalize about 35 million people labeled Canadians. Why on earth do we still generalize about more than 1 billion people labeled disabled? And yes, I am

Canadian. And listen hard to the disabled graduates new brain reaction? Why do you need a business case for treating me properly? Business Disability International wants to start a brand new conversation which drives a global revolution, as disabled and not yet disabled come together to ask - how do we replace the old brain with the new? And in case you're wondering who this Canadian is, I'm Susan Scott Parker, Chief Executive of Business Disability International.

03:48
Hello everyone, and welcome to "How do you lip read a robot?" It all started when I read an article which quoted a recruitment technology company saying how proud they were that their standardized assessment process removed human bias because every candidate would be treated exactly the same. I was startled. Surely everybody knows that the essence of treating candidates with disabilities fairly is to flex the process to make reasonable adjustments or accommodations in order to create that level playing field. It's called equal opportunities. Imagine asking the late great Stephen Hawking to climb stairs to the interview. Just because the standard process said everybody has to use the stairs. Is there anyone watching who would think it only right and proper to require the next Stephen Hawking to get up those downstairs just because everyone else is required to do so? I don't think so. So and by the way, if you're wondering, I'm a Canadian living in the UK, where I founded the world's first business disability network. And I set up and co-chaired the first Accessible Technology Task Force, we brought chief information officers and chief technology officers together to try to ensure that technology really did liberate everyone's potential. I now advise organizations like the United Nations, Global Business and Disability Network, and Valuable 500, mobilizing senior leadership, addressing the disability equality crisis across the multinational business community worldwide. However, my theme today, coming back to Mr. Hawking, is AI HR technology is the new stairs.

05:48
So my story started with questions. Did this article I read really matter? Was this particular recruitment tool, just a one-off, could it be ignored? Who would ever use it? But it immediately became very clear that for many recruiters, their primary task given the flood of candidates applying online is to discard as many as possible as quickly and as cheaply as possible. It also became immediately clear that neither developers nor their HR customers, despite all the talk about bias, were asking about disability bias. No one was asking about the impact of these tools on people with disabilities. The need for fair, ethical and responsible treatment of the world's more than 1.3 billion persons with disabilities and the millions of us who will become disabled with time is not on the HR technology and AI agenda. And then as we looked ever more closely at this exploding marketplace, we realized that we were looking at AI powered HR tools, which have the potential to damage the life chances of hundreds of millions of persons with disabilities around the world. And all this even though, at least one in three humans will be disabled or close to someone who is. 15 to 20% of the world's population in any country will have a disability - the largest of any minority groups, and one we can all join at any time. Even though one in five women have a disability, and at least one in three people aged 50 to 64 will be disabled, regardless of ethnicity. And well, between 10 to 12% of any large workforce will have a disability and or a chronic health condition. And even though if we live to be 70, and most of us would like to, if we live to be 70, most of us will experience at least 10 years of disability, which is why I was so puzzled that people with disabilities were never ever referenced in the growing debate regarding ethical and responsible AI. In

fact, it seems that disabled people are so missing from this debate that no one has even noticed they aren't there. However, I would remind you of a rather basic fact, ethics for some is not ethical. So if I could just illustrate some of what we're up against. I was told that a tool designed specifically to detect AI bias can't detect a disability bias because and I quote, "they aren't in the database". And given disabled people are always at least twice as unlikely to be in work, and given most HR technology data focuses on people in work. Well, I guess it's safe to say that if you aren't in that database, you never will be.

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And if we look at the actual assessment tools, neither their content, their usability, their accessibility, nor indeed their accuracy has ever been validated or tested on job seekers or employees with disabilities. So just to consider, we have these video interviews which assess the extent to which your nonverbals, your word choice, your voice pattern, eye movement, facial expression, etc, etc. match those of the ideal employee who's doing the job they assert they're seeking to fill. But they ignore the impact of disability on a candidate's nonverbals, on their word choice, eye movement, ability to smile, etc. But you're a math graduate who whispers because of your disability, or you lost your hearing as an adult and your voice pattern is unusual. Or you struggle to hear a Zune voice without captions, but the ideal employee doesn't use captions, so neither does the video interview. Applicants speaking English as a second language also like captions by the way. My real favorite is the virtual reality test. This one drops you into an ancient Egyptian tomb to test your problem solving skills. Now, my friend says he dreams in his wheelchair. He's used a wheelchair since he was a little boy. I'm afraid that virtual reality test isn't going to let him use his wheelchair as he navigates his way out of that tomb. I can't even begin to imagine what it would be like for him to find his avatar or himself walking after to so many years. And the increasingly popular gamified assessments, well how many have actually been checked to ensure that the controls can be used by people with limited dexterity or have only one hand or colorblindness? Did you see the post on LinkedIn from the young man who has disability, Tourette syndrome, causes his hands to twitch a bit. He wrote about how it was almost impossible, it took him more than 100 emails and calls to try to find any human recruiters to talk to, so that he could explain that he could do the job he just couldn't play the game. Now I really have to stress this is not just about bias. There is a crucial but unrecognized difference between biased data and the unfair treatment triggered by how the automated process actually works in practice. I'm tempted to say things like well bias is what we think and discrimination is what we do. I mean we certainly do say impairment is what happens to you, for example you have a speech impairment. Disability is what we do to you. We insist you first get interviewed on a phone and then hang up when you can't be immediately understood. Another one of my favorites has to be the Swedish robot interview. Its or should I say her website claims they have eliminated human bias regarding race and gender - there's no reference to disability bias but then there never is, so it's curious that humans have developed this and yet can claim that it's free of human bias. Mind you this bias-free creature is in fact a white female robot but that's another story. So let's think this through. How does this robot deal with a job candidate who could sail through if he could lip read the robot or use a sign language interpreter? Has the robot been taught not to interview the interpreter? I wonder. How does the robot adapt for the excellent candidate who stammers but needs slightly more than the standard, very standard, 20 minutes? Or someone who could do the job but because of their intellectual disability needs the robot to turn the standard questions into shorter questions or even using

different language. So I mean that really reminds me, I once met and very enlightened HR director working at a hospital. She was told that one potential and qualified candidate couldn't fill in the form as he had an intellectual disability so she invited him to the interview anyway. She put a rosebush in her offic,e she handed him the garden shears and asked him to trim the roses. He got the job because he could do the job and because she knew the job didn't require him to fill in forms. Just imagine what happens to that applicant when the hospital HR team decides it's all easier online. How does someone with should we call it a non-standard face - they had facial disfigurement acid burns, Down syndrome, someone with visual impairment where their eyes look a bit different, how do they get through a video interview when their face doesn't even register as a face? Or the person who doesn't make eye contact because they are physically unable to do so? Or someone who doesn't smile because they can't? Standardized, let's call them what they are, rigid recruitment processes are by definition inherently discriminatory especially when they're standardized by a predictable routine implementation of an Ai tool.

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I want to stress that AI powered unfairness does not only threaten the life chances of those among us with disabilities. I would urge every viewer to check out the recent experiment run by a team of very clever reporters at the german public broadcaster BR in Munich. Now their experiment had nothing to do with disability but as you listen I think you'll see the connection. They brought actors in who were given the same script and coached to always use the same expressions, they all had the same CV throughout the process. They were brought into a studio to test some software promising to identify personality traits of candidates based on a short video interview claiming that it was a much more objective and faster process than leaving it to old-fashioned human beings. The tool they evaluated is on the market, I mean this is scary but true. Their website states they work with big brands like Lufthansa, BMG. So how does this all start? Well the system claims to analyze your personality in less than the 60 seconds you are given to introduce yourself. They analyze your personality using what they call the ocean model, they're looking for openness, conscientiousness, extraversion, agreeableness and neuroticism, and now you don't have to be disabled for them to get this very wrong. One actress, again I stress using the same expression, the same words, the same CV, all she did differently in the second interview after her first assessment was to put on some glasses - it knocked 10 points off conscientiousness. Then, same actress, same words, same CV, she put on a headscarf. Her personality changes yet again from her first assessment without the headscarf, as she becomes more open, more conscientious and less neurotic by nearly 20 points. And almost best of all, when they added a painting, and slightly different wallpaper in what the camera could see behind the job seeker, the artificial intelligence apparently reacts and shifts your personality scores up by 15 points. Now, the developer says that they factor in how a human recruiter would have responded to the impression the candidate creates. I mean, this is fascinating. Do these developers really believe that recruiters in Germany adore headscarves? Are we to believe that wearing headscarves would give you an unfair advantage? Maybe if you're a woman, mind once this research becomes widely known I suspect, however, that everybody would wear headscarves at the interview, which presumably would even up the odds that the headscarf wearers would be unfairly advantaged. I just asked you to imagine the impact on your personality, as scored by this tool. If you've had a stroke, and your smile is unusual, imagine you have ADHD and your nonverbals are affected. Imagine you kind of hunched over because of your arthritis. Or instead of that painting behind your wall, the camera gets a glimpse of your wheelchair on this side or your crutches here. I stress if this tool works like this for non-disabled, just imagine, just imagine.

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So who is responsible for ensuring equal opportunities, for ensuring dignified and equitable access and fair treatment at every step of the way for every applicant? Well, developers say it's the employers responsibility. The employer says that our supplier says that this inflexible process is unbiased precisely because it is inflexible and standardized. Well, if they even think in HR of disability at all. Time and time again, neither the buyer nor the developer, make it possible at any stage for the applicant to request equal opportunities, for the applicant to request the adjustments that enable the fair treatment, that they by law in many countries have fully the right to expect. Online recruitment in its own right is notoriously unfair to many candidates, that's just online with or without any artificial intelligence involved. But imagine what happens when these badly designed AI tools are inserted into an already rigid, cumbersome, if not outright discriminatory online process. I tested an online recruitment process, there was no artificial intelligence elements so far, thank heavens, but I tested a recruitment process for a corporation they recruit in some more than 100 countries, all online, of course. I was pleased to see fairly early in, though not on the first page, that there was a button that said accessibility in the toolbar. So naturally, I clicked it. And up came the message, we regret we are unable to make this accessible in any way, and of course, not sad is and that's okay.

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And then we have the fourth dimension to this problem. So if I can just take us back a bit. One, we have disability bias in the data. Two, we have standardized processes that trigger unfair treatment. Three, we have HR buyers, diversity managers, and AI developers who do not understand disability discrimination. And in addition, we have developers who ignore the impact of future employer behavior on the tool's ability to accurately assess a given candidate who has a disability. To illustrate, how can an AI tool accurately assess a blind cybersecurity expert if it fails to ask, will the employer let this candidate use the tools she needs when she starts the job, whether it's voice activated software, screen readers, whatever, she cannot be accurately assessed for that cyber security job unless one, unless the AI tool recognizes that blindness on its own need not preclude high performance. Unless the tool does not assume that successful candidates will make eye contact in the standard way. Unless the assessment takes into account that the employer will or will not make the reasonable adjustments she needs if she is to succeed on the job. And finally, if the process does not stigmatize the candidate, who is regarded by the AI assessment, as so non-standard, that an expensive human assessor has to be brought into play. I call that the stigma bell ringing in the process. Now, if developers and their customers could just pick up on a particularly crucial message. AI tools that work for extreme users will work better for everyone. If women and girls with disabilities, remember one in five women have a disability, if women and girls with disabilities can be accurately and fairly assessed on their potential, it will be much easier for every woman to get a job on merit. If the disabled person from an ethnic minority can get through on their capability, well, anyone can get through on their capability, you get my point.

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So a community of thought leaders has come together determined to get human reality onto the responsible AI agenda. I stress we are just talking about human reality. And so this brings together the Oxford Brookes University Institute for Ethical AI, Goldsmiths College at the University of London, IBM globally, Simmons & Simmons the global law firm, the Australian Employers Network on Disability, the European Disability Forum, the World Bank, the World Economic Forum's Future of Work Initiative, and of course, the Center for Responsible AI here at New York University, led by the indomitable Julia Stoyanovich. So the good news is we are beginning to make progress. The first such white paper, called "Recruitment AI has a Disability Problem" by Selin Nugent focuses primarily on the risks to employers and job seekers. It was published recently by Oxford Brookes University's Institute for Ethical AI, who welcome comments, feedback, proposals to improve. And and let's be clear, they're spotting the procurement challenge here, which is very real. What questions must any employer ask before purchasing this high-risk technology? IBM recently published a second paper, "Designing AI Applications to Treat People with Disabilities Fairly: Six Steps to Fairness." And again, welcome comments, and suggestions. Simmons & Simmons who of course, are an employment law firm, continue to work with us as we look at what HR requires if they are to minimize the risks as buyers of this technology. So this is the buyer beware section of our work.

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And regulators are indeed beginning to pay attention, at least on grounds of race and gender. The key question for us all in terms of regulation, to me is captured by what is the difference between an AI tool that stops millions from competing fairly for work, and the teddy bear with button eyes that a child might swallow? Well the answer is the consumer protection regulator pulls the toy off the market. But the suppliers of the AI tool are not legally required to prove that their products are safe. However, regulators are beginning to consider adopting some of the principles that shape consumer protection legislation. We see the UN's committee monitoring the Convention on the Rights of Persons with Disabilities is for the first time drawing to the attention of the 181 countries that have ratified that UN Convention, they're drawing their attention to the risks and threats posed to people with disabilities by AI powered HR. There's a bill proposed by New York City which would require companies to disclose to candidates when they've been assessed with the help of an algorithm. And it is suggesting that companies that sell the software be required to do an annual audit to prove that they are not discriminatory. And I would just stress that this marketplace is growing. One industry analyst reports is currently tracking more than 1500 HR tech companies. All of this in the context of a wider trend that has more than a dozen jurisdictions in the United States alone, wanting to place legal constraints on the algorithms that shape life-changing decisions, like no, you didn't get a job. I would certainly argue that developers should be required by law to prove their AI tools are safe for disabled and other disadvantaged job seekers before they can put them on the market. We need regulators to catch up on this fast moving technology and to augment the traditional equality based legislation, which requires an individual to prove they've been treated badly by in this instance, an algorithm to adopting this anticipatory consumer protection approach. Indeed I'm told lawyers in America are eagerly awaiting the flood of litigation as candidates, and certainly not just those with disabilities, seek to prove they were treated badly by a recruitment algorithm. And not least, we see distinguished universities like New York, joining us and determine that the next generation of AI and robotics creators ground their work in a much more accurate and comprehensive understanding of what it means to be human. In conclusion, our challenge stems from the interplay between poorly trained HR professionals on the one hand, who are purchasing technology developed by developers who for some reason, do not ground their science in an understanding of human reality or the human condition. There must be a PhD research paper in there somewhere. Why do such clever people miss this? The best algorithm imaginable will not compensate for the policy which refuses to flex so that a non-standard candidate can be fairly evaluated. However, tolerating a rigid three minute rule for the video interview is ultimately the responsibility of both the employer and the developer. And ultimately, I believe it will be the responsibility of both to ensure that the data that sits at the heart of these AI tools more accurately reflects the human condition. Both the buyer and the HR tech suppliers have a lot of urgent homework to do. And we must also ask that HR ensure that their screening tools are not given the authority to discard candidates without human oversight, and that AI developers capture and share data regarding the candidates who are discarded. Almost all the data that's being fed into these systems at the moment is on people in work. What if we were to look at the individuals who did not get the job to help us determine if any patterns emerge that might highlight disability related and other discrimination? If 90% of applicants who failed to make it to the shortlist scored badly on eye contact or on voice pattern, for example, might that warrant disability related discrimination inquiries? Thankfully, we have in disability the perfect discrimination litmus test, we know that systems that work for extreme users work better for everyone. We can't say often enough, a recruitment assessment that enables someone who is deaf or blind, dyslexic, autistic, facially disfigured, mobility impaired, speech impaired, has a mental health condition, colorblind and and if the process enables them to be recruited on the basis of capability, and potential, it will make it easier for everyone to compete on the basis of their potential contribution. So that even a not yet disabled but Canadian woman can get through. Thank you for your kind attention. And thank you to Professor Julia Stoyanovich for her kind invitation to contribute to your vitally important program. Please don't hesitate to get in touch.